

# Project 5: Containerizing application and scanning its Docker image with DTR-Deploy a spring boot application on Docker for a retail company with multiple web applications.

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## 1. Download the retail project

```
Windows PowerShell
PS F:\SEM_7\my-git-practice\Day 5\retail-company> curl.exe https://start.spring.io/starter.zip -d type=maven-project
-d dependencies=web -d javaVersion=17 -d name=retail-api -d artifactId=retail-api -o retail-api.zip
% Total    % Received % Xferd  Average Speed   Time    Time     Current
           Dload  Upload   Total   Spent    Left     Speed
100 15110 100 15022 100    88 12934    75  00:01  00:01    0
PS F:\SEM_7\my-git-practice\Day 5\retail-company> tar -xf retail-api.zip
PS F:\SEM_7\my-git-practice\Day 5\retail-company> dir

Directory: F:\SEM_7\my-git-practice\Day 5\retail-company

Mode                LastWriteTime         Length Name
----                -
d-----          09-08-2026         10:09         .mvn
d-----          09-08-2026          09:52        gradle
d-----          09-08-2026         10:09         src
-a-----          09-08-2026         10:09          38 .gitattributes
-a-----          09-08-2026         10:09        394 .gitignore
-a-----          09-08-2026         10:00        218 .project
-a-----          09-08-2026          09:52        555 build.gradle
-a-----          09-08-2026          09:52       8654 gradlew
-a-----          09-08-2026          09:52       2846 gradlew.bat
-a-----          09-08-2026         10:09       1444 HELP.md
-a-----          09-08-2026         10:09      11790 mvnw
-a-----          09-08-2026         10:09       8292 mvnw.cmd
-a-----          09-08-2026         10:09       1371 pom.xml
-a-----          09-08-2026         10:10      15022 retail-api.zip
-a-----          09-08-2026          09:52         32 settings.gradle

Windows PowerShell
PS F:\SEM_7\my-git-practice\Day 5\retail-company> docker build -t retail-api:v1 .
[+] Building 123.2s (16/16) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile                  0.1s
=> => transferring dockerfile: 623B                                0.0s
=> [internal] load metadata for docker.io/library/eclipse-temurin:17-jre 4.0s
=> [internal] load metadata for docker.io/library/maven:3.9.6-eclipse-temurin-17 4.0s
=> [auth] library/maven:pull token for registry-1.docker.io        0.0s
=> [auth] library/eclipse-temurin:pull token for registry-1.docker.io 0.0s
=> [internal] load .dockerignore                                    0.0s
=> => transferring context: 2B                                       0.0s
=> [build 1/5] FROM docker.io/library/maven:3.9.6-eclipse-temurin-17@sha256:29a1658b1f3078e07c2b17f7b519b45e 71.1s
=> => resolve docker.io/library/maven:3.9.6-eclipse-temurin-17@sha256:29a1658b1f3078e07c2b17f7b519b45eb47f65d 0.0s
=> => sha256:fc06d68d71ba375a9ceed165c0c04b09b1a8cc193146888663b005660fc25013 156B / 156B 0.6s
=> => sha256:4712dfa85971124b8c3507fbba4c02441e5754f1517162926e600afd3fc5404b 358B / 358B 0.7s
=> => sha256:45f480637770dbb740623c0c7f827d44bbacc9204faf3d07b4135d54a2bee043 9.48MB / 9.48MB 14.4s
=> => sha256:58c3491a14ebc973960cbda9f2eeaa22537785a3cb07e81487bd1115bad4a8278 0B / 852B 101.5s
=> => sha256:67f99c2668af1d0eced6d61dad057d7beffe543a8e8370d5c0d3fe8682e059a3 19.00MB / 19.00MB 19.2s
=> => sha256:a3ba11f7aaadedad962216812fe84aee9061aeabac6932f0274a1d204eb96e8e8 734B / 734B 0.9s
=> => sha256:c97285723537ab7921fca4d081c256e501adfdad8992d04637992075f4cea392 173B / 173B 0.6s
=> => sha256:d59fd278c1b4ffe1727be6ccba42125e8f4db57e660e795cce19889d2c776457 145.10MB / 145.10MB 46.2s
=> => sha256:5e5d1bcc5440d3a24f4a620704b9e687b4163c6c872fcc8e812e200c9bbac58 17.46MB / 17.46MB 25.5s
=> => sha256:4a023cab5400feb5c1ab725beb8345d5db0e3200314004b56677a5eee2e8c86cf 30.44MB / 30.44MB 19.4s
=> => extracting sha256:4a023cab5400feb5c1ab725beb8345d5db0e3200314004b56677a5eee2e8c86cf 1.2s
=> => extracting sha256:5e5d1bcc5440d3a24f4a620704b9e687b4163c6c872fcc8e812e200c9bbac58 1.4s
=> => extracting sha256:d59fd278c1b4ffe1727be6ccba42125e8f4db57e660e795cce19889d2c776457 2.3s
=> => extracting sha256:c97285723537ab7921fca4d081c256e501adfdad8992d04637992075f4cea392 0.0s
=> => extracting sha256:a3ba11f7aaadedad962216812fe84aee9061aeabac6932f0274a1d204eb96e8e8 0.0s
=> => extracting sha256:67f99c2668af1d0eced6d61dad057d7beffe543a8e8370d5c0d3fe8682e059a3 0.8s
=> => extracting sha256:45f480637770dbb740623c0c7f827d44bbacc9204faf3d07b4135d54a2bee043 0.2s
=> => extracting sha256:58c3491a14ebc973960cbda9f2eeaa22537785a3cb07e81487bd1115bad4a8278 0.0s
=> => extracting sha256:4712dfa85971124b8c3507fbba4c02441e5754f1517162926e600afd3fc5404b 0.0s
```

```
Windows PowerShell
=> => extracting sha256:4712dfa85971124b8c3507fbba4c02441e5754f1517162926e600afd3fc5404b 0.0s
=> => extracting sha256:fc06d68d71ba375a9ceed165c0c04b09b1a8cc193146888663b005660fc25013 0.0s
=> [stage-1 1/3] FROM docker.io/library/eclipse-temurin:17-jre@sha256:e4f018a55645ad204892e44eb35437518d7e10 43.4s
=> => resolve docker.io/library/eclipse-temurin:17-jre@sha256:e4f018a55645ad204892e44eb35437518d7e108ba2a2dce 0.0s
=> => sha256:26df7c06c61d8c89938b275761188d37d909096584bae567db05ad2959e406df 2.28kB / 2.28kB 0.4s
=> => sha256:08758b58d308b0203443825c9cdee0fa66e88ea686000d8fc0072e19cfb109e 159B / 159B 0.8s
=> => sha256:617772c7d19b9f9c4141bc73b37e6f4c5e4e69729f6881a73d0dc23676dbbd60 41.57MB / 41.57MB 19.6s
=> => sha256:5900e0e4b3f7100211678e03f8e8d08acb783c4f6ca348a05fb07e0d624c58b1 29.39MB / 29.39MB 21.8s
=> => sha256:dd424020a6d5ec0fafd4f51e23e6d19301d3297d4f97af4636caf6d9e8522a47 47.57MB / 47.57MB 40.2s
=> => sha256:a7fb98a8eddd1a70f99f205cb4b66e830a3e8ef4798f5636ece054a567b650de 389B / 389B 0.6s
=> => extracting sha256:617772c7d19b9f9c4141bc73b37e6f4c5e4e69729f6881a73d0dc23676dbbd60 2.7s
=> => extracting sha256:a7fb98a8eddd1a70f99f205cb4b66e830a3e8ef4798f5636ece054a567b650de 0.0s
=> => extracting sha256:5900e0e4b3f7100211678e03f8e8d08acb783c4f6ca348a05fb07e0d624c58b1 1.3s
=> => extracting sha256:dd424020a6d5ec0fafd4f51e23e6d19301d3297d4f97af4636caf6d9e8522a47 2.2s
=> => extracting sha256:08758b58d308b0203443825c9cdee0fa66e88ea686000d8fc0072e19cfb109e 0.1s
=> => extracting sha256:26df7c06c61d8c89938b275761188d37d909096584bae567db05ad2959e406df 0.1s
=> [internal] load build context 0.1s
=> => transferring context: 2.79kB 0.0s
=> [stage-1 2/3] WORKDIR /app 0.6s
=> [build 2/5] WORKDIR /app 0.8s
=> [build 3/5] COPY pom.xml . 0.1s
=> [build 4/5] COPY src ./src 0.1s
=> [build 5/5] RUN mvn clean package -DskipTests 44.8s
=> [stage-1 3/3] COPY --from=build /app/target/*.jar app.jar 0.1s
=> exporting to image 1.5s
=> => exporting layers 1.0s
=> => exporting manifest sha256:7130c22fe75bb92f2a4d6056de41fc501a143cd20b84746bc328127cb465ff1d 0.0s
=> => exporting config sha256:900af96672110706dadcd68c7a64cf01f8fc26b2a104d72fd6026c1162d74509f 0.0s
=> => exporting attestation manifest sha256:2080e244a1f5f3abeb2d8efbd094904fc2b7f9fd7235129f4184cf0f448c28d3 0.0s
=> => exporting manifest list sha256:1737a79d0a10ee48ca4230a7eeb16d3616ae6133feb96197a8762a97fdf3979b 0.0s
=> => naming to docker.io/library/retail-api:v1 0.0s

=> => naming to docker.io/library/retail-api:v1 0.0s
=> => unpacking to docker.io/library/retail-api:v1 0.3s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/qu55j52v92528geau6pg9rfa1
PS F:\SEM_7\my-git-practice\Day 5\retail-company>
```

docker desktop

PERSONAL

Ctrl+K

A

Gordon

Containers

Images

Logs

Volumes

Kubernetes

Builds

Models

Docker Hub

MCP Toolkit BETA

Extensions

Builds

[Give feedback](#)

Selected builder

desktop-linux

Import builds

Builder settings

Build history

Active builds

Show only my builds

| <input type="checkbox"/> | Name                   | ID     | Builder | Durati... | Created       | Build Clo... | Actions |
|--------------------------|------------------------|--------|---------|-----------|---------------|--------------|---------|
| <input type="checkbox"/> | ✓ Day 5/retail-company | qu55j5 | default | 2m 02s    | 5 minutes ago | N/A          |         |
| <input type="checkbox"/> | ✗ Day 5/retail-company | vzm4dv | default | 0.2s      | 7 minutes ago | N/A          |         |
| <input type="checkbox"/> | ✓ flask-postgres-app   | 099e8l | default | 24.8s     | 18 days ago   | N/A          |         |
| <input type="checkbox"/> | ✓ my-node-app          | 4pgwwu | default | 41.1s     | 18 days ago   | N/A          |         |

Rows per page: 10 1-4 of 4

Upgrade plan

Engine running

Kubernetes failed to start RAM 2.42 GB CPU 1.25% Disk: 9.60 GB used (limit 1006.85 GB)

Terminal v4.81.0

## 2. Security scanning

```
PS F:\SEM_7\my-git-practice\Day 5\retail-company> docker scout quickview retail-api:v1
i New version 1.24.0 available (installed version is 1.22.0) at https://github.com/docker/scout-cli
v Image stored for indexing
v Indexed 256 packages
! failed to delete temporary image archive C:\Users\Aarya\AppData\Local\Temp\docker-scout\sha256\1737a79d0a10ee48ca4230a7eeb16d3616ae6133feb96197a8762a97fdf3979b\cf93190f-80ff-402d-aa68-f30ca0c88173: unlinkat C:\Users\Aarya\AppData\Local\Temp\docker-scout\sha256\1737a79d0a10ee48ca4230a7eeb16d3616ae6133feb96197a8762a97fdf3979b\cf93190f-80ff-402d-aa68-f30ca0c88173: The process cannot access the file because it is being used by another process.
i Base image was auto-detected. To get more accurate results, build images with max-mode provenance attestations.
Review docs.docker.com for more information.
```

|                    |                        |    |    |    |    |
|--------------------|------------------------|----|----|----|----|
| Target             | retail-api:v1          | 0C | 0H | 1M | 0L |
| digest             | 1737a79d0a10           |    |    |    |    |
| Base image         | eclipse-temurin:17-jre | 0C | 0H | 0M | 0L |
| Updated base image | eclipse-temurin:26-jre | 0C | 0H | 0M | 0L |

### What's next:

View vulnerabilities → `docker scout cves retail-api:v1`  
View base image update recommendations → `docker scout recommendations retail-api:v1`  
Include policy results in your quickview by supplying an organization → `docker scout quickview retail-api:v1 --org <organization>`

```
PS F:\SEM_7\my-git-practice\Day 5\retail-company> |
```

```
Windows PowerShell
PS F:\SEM_7\my-git-practice\Day 5\retail-company> docker scout cves retail-api:v1
i New version 1.24.0 available (installed version is 1.22.0) at https://github.com/docker/scout-cli
v SBOM of image already cached, 256 packages indexed
x Detected 1 vulnerable package with 1 vulnerability

## Overview
```

|                 | Analyzed Image |
|-----------------|----------------|
| Target          | retail-api:v1  |
| digest          | 1737a79d0a10   |
| platform        | linux/amd64    |
| vulnerabilities | 0C 0H 1M 0L    |
| size            | 136 MB         |
| packages        | 256            |

```
## Packages and Vulnerabilities

0C 0H 1M 0L tools.jackson.core/jackson-databind 3.1.4
pkg:maven/tools.jackson.core/jackson-databind@3.1.4

x MEDIUM CVE-2026-59889 [Incorrect Authorization]
https://scout.docker.com/v/CVE-2026-59889
Affected range : >=3.0.0
                  : <=3.1.4
Fixed version   : 3.1.5
CVSS Score      : 6.5
CVSS Vector     : CVSS:3.1/AV:N/AC:L/PR:L/UI:N/S:U/C:N/I:H/A:N
```

### 1 vulnerability found in 1 package

|          |   |
|----------|---|
| CRITICAL | 0 |
| HIGH     | 0 |
| MEDIUM   | 1 |
| LOW      | 0 |

### What's next:

View base image update recommendations → `docker scout recommendations retail-api:v1`

```
PS F:\SEM_7\my-git-practice\Day 5\retail-company> |
```

## 3. Running Docker

```
PS F:\SEM_7\my-git-practice\Day 5\retail-company> docker run -d -p 8080:8080 --name retail-app retail-api:v1
8acb0e15c50f1583e96211c07221b1c6e2a619338e2393e486774af9297b5b26
PS F:\SEM_7\my-git-practice\Day 5\retail-company> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
8acb0e15c50f   retail-api:v1  "java -jar app.jar"     2 minutes ago Up 2 minutes   0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp
PS F:\SEM_7\my-git-practice\Day 5\retail-company> |
```